Twin Counting Units: A Model-Free Test for Exact-Tie Anomalies in Election Returns, with Evidence from Two Korean Elections

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Abstract

In the 2026 Korean nationwide local elections, nine "twin" pairs of counting units — distinct (neighborhood × vote-type) units whose two leading candidates received exactly identical vote counts — reignited public election-fraud controversies. Using the complete official returns (17 provinces, ~7,000 counting units), we measure how surprising this is. First, a multinomial–binomial generative model with per-unit observed parameters yields an expected 3.23 twins and $P(\geq 9) = 0.64\%$. Second, we prove that such plug-in (parametric-bootstrap) procedures underestimate the collision probability of truly identical-parameter pairs by a factor of $1/\sqrt{2}$ per matched dimension — a noise double-counting bias — and propose as a complement a **model-free sideband estimator**, the bump-hunt analogue for exact ties, which extrapolates lattice-shell near-pair counts to distance zero. The sideband estimate, 3.47 ± 0.18 expected twins with $P(\geq 9) = 0.94\%$ [0.49, 1.66], converges with the generative model, as does an unconditional distribution-based (ab-initio) simulation (~1%). Third, a pre-specified replication on the 2025 presidential election shows no excess (7 observed vs. 6.02 expected, P = 39.7%), with near-pair density ratios and vote-type composition matching the model. Fourth, type decomposition localizes the local-election anomaly: all five Gwangju–Jeonnam twins fall in the early-voting pair type with the *smallest* expectation (conditional $P \approx 0.2\%$, post hoc), so the anomaly is confined to "local elections × Gwangju–Jeonnam × early voting × exact ties." We conclude that (i) twins are a baseline natural phenomenon; (ii) the 2026 excess is a ~1% tail event — too rare for "entirely natural," far too common for "evidence of fraud"; and (iii) the residual identity of the excess (chance vs. a non-statistical cause) is unidentifiable from a single election. As a by-product we show that the key assumption of the public analysis that sparked the debate — a 1% "similar-pair" rate — is about 80× too large (measured: 0.012%).

Keywords: election forensics, exact ties, birthday problem, parametric bootstrap, bump hunt, post-hoc selection

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1. Introduction

1.1 The controversy

Immediately after the 9th Korean nationwide local elections (June 3, 2026), it was noticed that in the Incheon mayoral race the two leading candidates' early-voting counts in two neighborhoods of Songdo (Yeonsu-gu) coincided exactly at (3,030, 1,440), and that the South Jeolla gubernatorial race contained five such "twin" pairs. Fraud allegations followed. A statistician responded with a public analysis arguing, via a coin-flipping model and birthday-problem logic, that the coincidences are "mathematically natural" [8]; rebuttals and counter-rebuttals ensued. Both sides argued from weak quantitative ground: the suspicious side never computed the base rate of exact ties, while the rebuttal substituted an assumption for the load-bearing parameter of its own argument (a "1% similar-pair rate").

1.2 Research question and contributions

Our question is simple: **is the observed number of twins compatible with the base rate predicted by a smooth stochastic model?** We make four contributions.

1. **Measurement.** Using the complete official returns we measure both the observed twin counts and their expectation — by model-dependent and model-free routes — replacing the assumptions of the public debate with measured quantities.
2. **A methodological warning.** We prove that plug-in simulations which redraw counts around observed parameters systematically underestimate exact-tie probabilities, and characterize when and by how much ($1/\sqrt{2}$ per dimension; Proposition 1).
3. **A model-free estimator.** We propose a sideband estimator of the expected number of exact ties — extrapolating lattice-shell counts of near pairs to distance zero — derive its unordered-pair normalization (Proposition 2), and validate it by simulation self-consistency.
4. **A replication design.** We pre-specify the prediction "if the excess is structural it must reappear, scaled up, in another election," and test it on the 2025 presidential election, where it is rejected. This partially resolves the circularity inherent in post-hoc analysis of a single election.

1.3 Related work

Statistical detection of electoral irregularities includes digit-based tests [1, 2], turnout–vote-share fingerprints [3], and integer-percentage / duplicate-value anomalies [4]. Our object — exact multivariate ties between *different* units — is kin to the duplicate analyses of [4], but differs in that (i) we treat two-dimensional collisions (both leading candidates simultaneously), and (ii) we estimate the expectation not from a model but from the data's own near-pair density, in direct analogy with the bump hunt of high-energy physics (interpolating the expected background in a signal region from sidebands). On the dangers of post-hoc hypothesis selection we follow [5].

2. Data

2.1 The 2026 local elections (June 3, 2026)

We collected the complete gubernatorial/mayoral returns for all 17 provinces from the National Election Commission's election statistics system (info.nec.go.kr) with an automated Playwright crawler. A **counting unit** is a neighborhood (eup/myeon/dong) $\times$ vote type {in-precinct early voting, election-day voting}; absentee, out-of-precinct early votes, and overseas votes are not broken down by neighborhood and are excluded. The two leading candidates (A, B) are identified per province by total votes. Gwangju and South Jeolla share the same two leading candidates (same race) and are merged into a single block, so that cross-province pairs count; all other provinces have distinct candidates, so only within-province pairs are meaningful. The final data comprise ~3,500 neighborhoods and ~7,100 counting units.

A convention of the source data matters: the "registered voters" of an early-voting unit equals its turnout (tautologically), and a neighborhood's total electorate equals the sum of its early-voting and election-day registered counts.

2.2 The 2025 presidential election (June 3, 2025)

We use the official polling-station-level CSV from the public data portal (169,749 rows), aggregating election-day polling stations to the neighborhood level to match the unit definition above. The presidential election has identical candidates nationwide, so the whole country forms a single block: 3,554 neighborhoods → 7,108 counting units, with ~2.5×10^7 comparable pairs. (A, B) = (Lee Jae-myung, Kim Moon-soo).

2.3 Observed twins

The local elections produced nine twins (Table 1): five in the Gwangju–Jeonnam block (including one cross-province pair) and one each in Incheon, North Gyeongsang, South Gyeongsang, and North Jeolla. The vote-type composition is six early · early, two day · day, and one mixed pair.

Table 1. The nine twins of the 2026 local elections

Block	Unit 1	Unit 2	A	B	Type
Incheon	Songdo 1-dong, Yeonsu	Songdo 2-dong, Yeonsu	3,030	1,440	early · early
Gwangju–Jeonnam	Songjeong 1-dong, Gwangsan	Geumsan-myeon, Goheung	1,401	120	early · early (cross)
Gwangju–Jeonnam	Bukha-myeon, Jangseong	Eomda-myeon, Hampyeong	606	57	early · early
Gwangju–Jeonnam	Samil-dong, Yeosu	Hauido-myeon, Sinan	506	42	early · early
Gwangju–Jeonnam	Iyang-myeon, Hwasun	Byeongyeong-myeon, Gangjin	444	46	early · early
Gwangju–Jeonnam	Nodong-myeon, Boseong	Palgeum-myeon, Sinan	356	42	early · early
N. Gyeongsang	Jeomgok-myeon, Uiseong	Changsu-myeon, Yeongdeok	456	109	day · day
S. Gyeongsang	Yangbo-myeon, Hadong	Wicheon-myeon, Geochang	385	180	day · day
N. Jeolla	Ungpo-myeon, Iksan	Seongsu-myeon, Jinan	186	150	early · day

The presidential election produced seven twins (six day · day, one mixed, zero early · early; §5).

2.4 Data integrity verification

For all 18 counting units composing the twins of Table 1 we performed two checks (`src/verify_twins.py`). (i) **Internal arithmetic consistency**: early + election-day = total (for electorate, ballots, A, and B separately), and A + B ≤ ballots — all units pass. (ii) **Cross-validation across independent retrievals**: the nine Jeonnam units were collected twice, by the automated crawler and in a separate manual crawl session; the two retrievals agree exactly on every unit. The data therefore faithfully reflect the published official returns, and the observed twins are not artifacts of our collection or transcription. Errors *upstream* of official publication (counting floor → reporting system) cannot be excluded by re-reading the published records; that is the domain of physical recounts (§7.2).

3. Methods

3.1 The canonical generative model

For neighborhood i, let N_i be its electorate, r_i^e its early-voting rate, r_i^b its election-day rate conditional on not voting early, and $p^t_{A,i}, p^t_{B,i}$ the leading candidates' vote shares by vote type $t \in \{e, b\}$ — all set to their observed per-neighborhood values. One realization is generated as

$$V_i^e \sim \mathrm{Bin}(N_i, r_i^e), \quad V_i^b \sim \mathrm{Bin}(N_i - V_i^e, r_i^b),$$
$$A_i^t \sim \mathrm{Bin}(V_i^t, p^t_{A,i}), \quad B_i^t \sim \mathrm{Bin}(V_i^t - A_i^t,\ p^t_{B,i}/(1 - p^t_{A,i})),$$

and we count unordered pairs of units agreeing simultaneously in (A, B). The only randomness is (i) who votes early / on election day / abstains (multinomial) and (ii) how ballots split (binomial); there are no free parameters. National distributions are obtained as sums over blocks simulated with independent seeds (R = $5{\times}10^4$–$2{\times}10^5$ per block).

Robustness variants — fixed vs. random turnout, sub-binomial dispersion $\varphi < 1$, population-draw parameters, finite-population corrections — do not change the results (§5.1 of the repository's research notes). We additionally run an **ab-initio variant** that deliberately severs all parameter correlations: only the number of neighborhoods is kept, and every realization redraws N_i from a fitted lognormal and all rates from fitted Beta distributions (method-of-moments). This provides an unconditional base rate to which the criticism "you conditioned on the observed values" does not apply.

3.2 The plug-in bias (noise double-counting)

The canonical model is a parametric bootstrap that plugs observed parameters in as truth. For exact ties this procedure carries a systematic bias.

Proposition 1. Let two units share identical true parameters, $X_i, X_j \sim \mathrm{Bin}(V, p)$ independently. Re-drawing around the observations, $X'_k \sim \mathrm{Bin}(V, X_k/V)$, the tie probability satisfies asymptotically

$$\frac{\mathbb{E}_{\text{data}}\, P(X_i' = X_j' \mid \text{data})}{P(X_i = X_j)} \;\longrightarrow\; \frac{1}{\sqrt{2}},$$

and $2^{-d/2}$ for simultaneous ties in d coordinates.

Proof sketch. Under the normal approximation $D = X_i - X_j \approx \mathcal{N}(0, 2\sigma^2)$ with $\sigma^2 = Vp(1-p)$, so $P(D = 0) \approx (4\pi\sigma^2)^{-1/2}$. The re-drawn difference is centered on the observed difference d: $D' \mid d \approx \mathcal{N}(d, 2\sigma^2)$, and d itself is distributed $\mathcal{N}(0, 2\sigma^2)$; marginally $D' \approx \mathcal{N}(0, 4\sigma^2)$ — the noise is convolved in twice. The ratio of densities at zero is $\sqrt{2\sigma^2/4\sigma^2} = 1/\sqrt{2}$; coordinates multiply under independence. ■

Numerical verification (exact binomial pmf inner products; V = 2,000, 50 identical pairs plus 100 heterogeneous units, 200 observation replicates): ratio 0.707 for identical pairs (theory $1/\sqrt{2}$), 0.985 (≈ 1) for heterogeneous pairs. Conversely, when the distribution of true parameter differences is locally flat at the noise scale the bias vanishes — its magnitude depends on unobservable structure (how many truly identical pairs exist) and is therefore uncorrectable within the model. This motivates the model-free estimator below.

3.3 The model-free sideband estimator

For each pair (i, j) define the Chebyshev distance $d_{ij} = \max(|A_i - A_j|, |B_i - B_j|)$ and the shell counts $n(d) = \#\{i < j : d_{ij} = d\}$. Twins are $n(0)$.

Proposition 2 (unordered-pair normalization). If the per-lattice-site density c of ordered pair differences is locally constant near the origin, then the L_∞ shell at distance $d \geq 1$ contains $8d$ lattice sites and the unordered-pair shell count has expectation $\mathbb{E}\, n(d) = 4dc$, while the origin has $\mathbb{E}\, n(0) = c/2$. Hence with $\rho(d) \equiv n(d)/(8d)$, $\mathbb{E}\, n(0) = \rho(0)$: **the $d \to 0$ extrapolation of $\rho(d)$ is itself the expected number of twins.**

Proof sketch. Unordered pairs identify $\pm v$; each class with $v \neq 0$ absorbs two lattice sites, giving $4d$ classes per shell, while the origin class absorbs the two ordered self-reflections, giving expectation $c/2$. Synthetic uniform check: $n(0) = 87.8$ vs. extrapolation 85.5 (theory 88.9). ■

We fit a linear (and, as a check, quadratic) trend to $\rho(d)$, $d \in [1, 15]$, by Poisson maximum likelihood, extrapolate to $\rho(0)$, obtain standard errors by Poisson resampling (300 replicates), and compute tail probabilities from Poisson($\hat{\rho}(0)$). **Self-validation**: applying the same extrapolation to the generative model's own simulated shells recovers the model's true $d = 0$ expectation (3.24 vs. 3.27 ± 0.01).

This estimator uses no generative assumptions — no binomiality, no turnout model, no dispersion parameter. Its only assumption is that the distribution of count differences between distinct units is smooth at the ±15-vote scale near the origin, which is precisely the null hypothesis under test. As a companion diagnostic we examine the observed-to-model shell ratio $r(d) = n_{\text{obs}}(d)/n_{\text{sim}}(d)$: a uniformly elevated, flat $r(d)$ would indicate global over-smoothing by the model (Proposition 1's bias at work), whereas $r(d) \approx 1$ for $d \geq 1$ with an isolated spike at $r(0)$ indicates an atom at exact equality.

3.4 Replication test and type decomposition

We pre-specify: *if the local-election excess reflects a structural mechanism of elections in general (model misspecification, reporting practice, demographic clustering), then the presidential election — whose expectation is larger — must show a proportionally larger absolute excess (*$6.0 \times 2.8 \approx 17$ *twins).* We also decompose expected twins by vote-type pair {early · early, day · day, mixed} and compare with the observed composition. To first order the expected density of exact ties is independent of the noise scale and is determined by the lattice packing density of unit means, $\propto 1/(\text{spread}_A \times \text{spread}_B)$; the decomposition therefore yields model and data predictions for *where* twins should occur.

4. Results I: the 2026 local elections

4.1 National base rate and significance

Table 2. Nationwide results (9 observed twins)

Estimation route	Expected twins	P(≥ 9)
Canonical generative model	3.23	0.64%
Sideband, linear extrapolation	3.47 ± 0.18	0.94% [0.49, 1.66]
Sideband, quadratic	3.64 ± 0.36	1.26% [0.33, 3.39]

For the Gwangju–Jeonnam block alone (5 observed): canonical expectation 1.47 with $P(\geq 5) = 1.74\%$; sideband 1.67 ± 0.12 with $P(\geq 5) = 2.75\%$; ab-initio (unconditional, per-province/pooled fits) expectation 1.26/1.33 with $P(\geq 5) = 0.99\%/1.26\%$. **The conditional, unconditional, and model-free routes all converge on the 1–3% band.**

4.2 Shell diagnostics: the excess is an atom at exact equality

Table 3. Nationwide near-pair shells: observed vs. model

d	0	1	2	3	4	5	6	8	10	12	15
Observed	9	35	48	72	130	146	135	198	238	315	362
Model	3.24	25.8	51.2	76.5	102.1	128.5	153.5	203.6	252.2	299.6	368.5
r(d)	**2.78**	1.36	0.94	0.94	1.27	1.14	0.88	0.97	0.94	1.05	0.98

Figure 1. Nationwide near-pair shells, 2026 local elections. Observed density $\rho(d)$ with Poisson error bars, generative model (grey), and the $d \geq 1$ sideband fit extrapolated to $d = 0$ (red). The gap between the nine observed exact ties (star) and the extrapolated 3.47 is the excess under test.

For $d \geq 2$ the ratio $r(d)$ is flat at 0.88–1.27 (within Poisson fluctuation): **the model reproduces the density of near-misses exactly.** Had the hidden population of identical-parameter pairs presupposed by Proposition 1's bias existed, r would have been elevated across the whole range (~2); it is not, so the canonical expectation 3.23 is trustworthy and its agreement with the sideband 3.47 is independent confirmation. The only departure is at $d = 0$: the excess is not

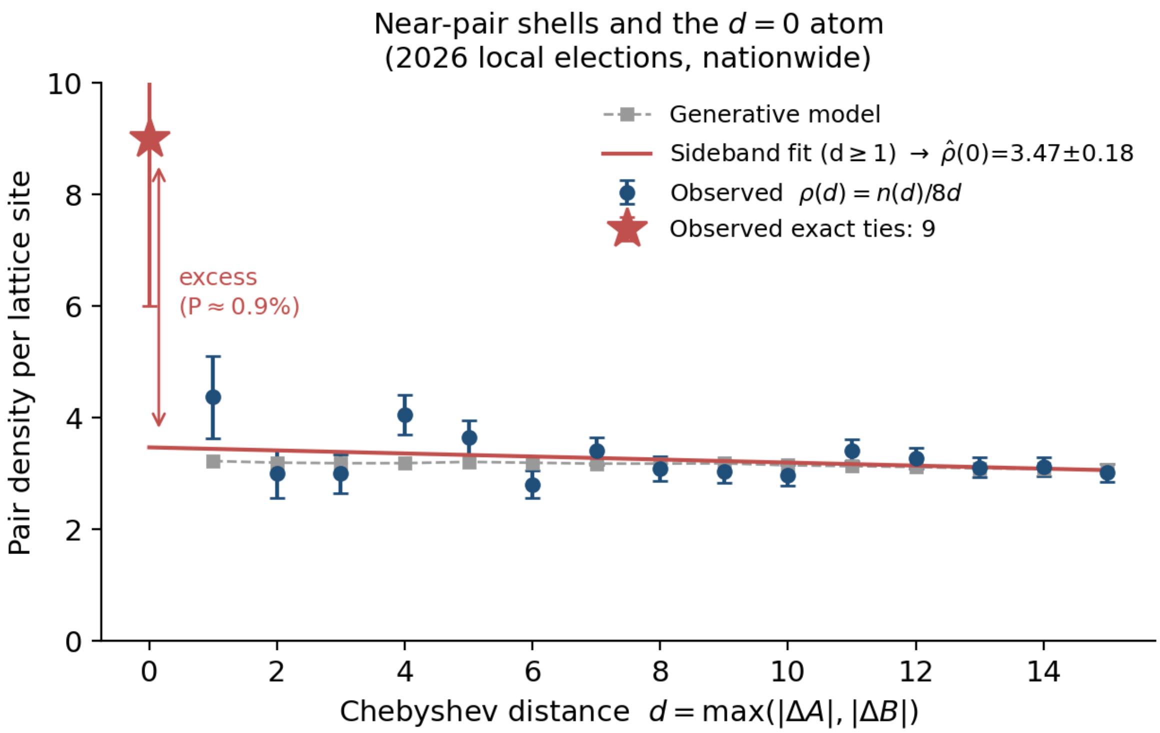

Figure 1: Figure 1

smooth clustering but an **atom at exact equality**. This excludes both directions of smooth explanation at once: a benign mechanism such as demographic similarity would also inflate $d = 1$–15 (no trace), and the over-smoothing critique of the model is refuted by $r(d \geq 1) \approx 1$.

4.3 Type decomposition: the excess sits in the least-expected type

The decomposition for the Gwangju–Jeonnam block (Table 4) reveals a fact opposite to intuition. Election-day units are *smaller* than early-voting units (median ballots 796 vs. 1,147); both the data's own near pairs and the model's expectation rank day · day pairs first. Yet all five observed twins are early · early — the type with the smallest expectation.

Table 4. Gwangju–Jeonnam type decomposition

	day · day	mixed	early · early
Model expected twins	0.58	0.48	**0.42**
Observed near pairs (d = 1–10)	265	198	174
Observed twins	0	0	**5**

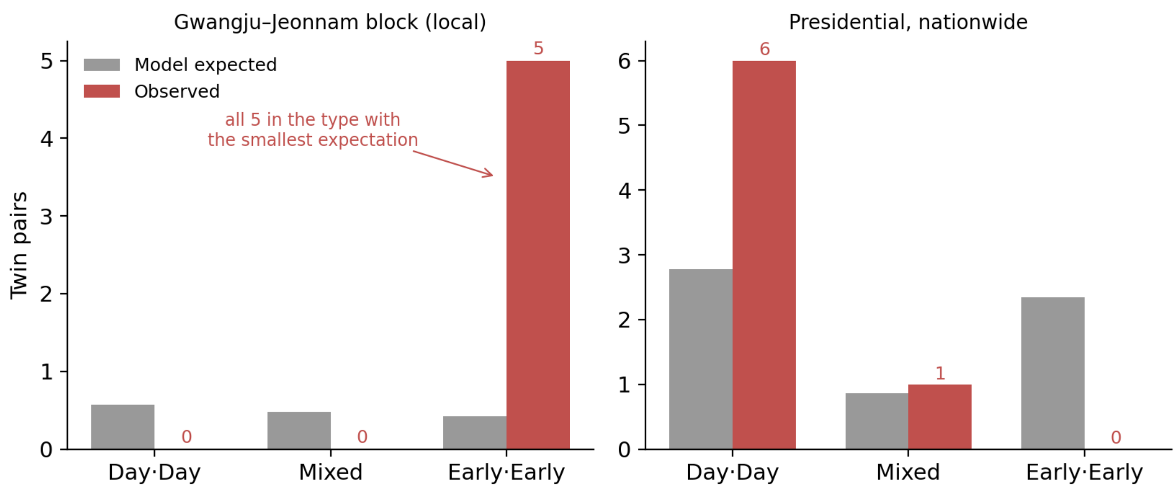

Figure 2: Figure 3

Figure 3. Expected (grey) vs. observed (red) twins by type. Left: Gwangju–Jeonnam (local) — all five observed twins fall in the least-expected type. Right: presidential — observed composition matches expectation.

Conditioning on the total, $P(\text{all 5 early} \cdot \text{early} \mid \text{type share } 28.4\%) = 0.284^5 \approx 0.19\%$ — an anomaly *additional* to the count excess (1.7%). It is a post-hoc statistic, selected after seeing the data, and should be discounted for the exploration degrees of freedom (three types, block choice); it remains hard to dismiss after that discount. In sum the anomaly is localized four-fold: **local elections** $\times$ **Gwangju–Jeonnam** $\times$ **early voting** $\times$ **exact equality**. The popular benign account — "landslide $\times$ small rural units, hence natural" — fails the type decomposition, since the same logic predicts a day · day majority.

5. Results II: replication — the 2025 presidential election

The identical pipeline applied to the nationwide single block (7,108 units):

Table 5. Presidential election (7 observed twins)

Estimation route	Expected	P(≥ 7)
Canonical generative model	6.02	39.7%
Sideband, linear extrapolation	6.44 ± 0.24	46.3%

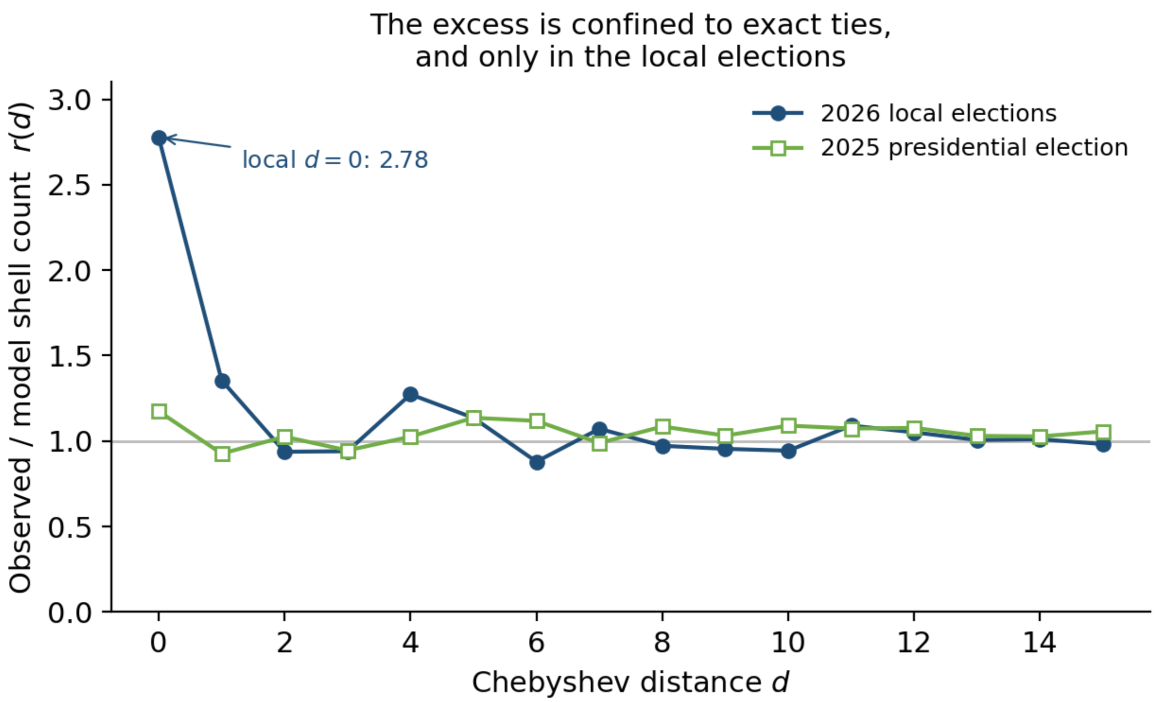

Figure 3: Figure 2

Figure 2. Shell ratio $r(d)$ for the two elections. The presidential election (green) is consistent with 1 everywhere including $d = 0$; the local elections (blue) depart only at $d = 0$ (2.78).

The ratio $r(d)$ is flat at 0.93–1.17 across the entire range including $d = 0$. The type composition (model expectation: day · day 2.78, early · early 2.35, mixed 0.86; observed: 6, 0, 1) is consistent with the model (P(zero early of 7 | 39%) ≈ 3%). The pre-specified prediction — ~17 twins under a structural mechanism — is decisively rejected ($P(\geq 17 \mid 6.0) \approx 10^{-4}$).

The reversal of type composition between the elections (early 6/9 → early 0/7) follows naturally from the packing-density principle: twins track not a particular vote type but *wherever counts bunch up small*. In Jeonnam's local race the landslide (B's share 8%) compressed the early-voting B-axis to 42–57 votes; in the presidential race the election-day units of rural Honam played that role. This refutes the popular notion of an early-voting-specific anomaly with data — while reaffirming that the localization of §4.3 (why the *least*-expected type?) cannot be explained by any property of elections in general.

6. Re-evaluating the public analysis

The public analysis [8] that sparked the debate used the decomposition (number of pairs) × (similar-pair rate) × (tie probability | similar). This skeleton is a hand-computed version of our sideband estimator and is structurally sound. The problem is the provenance of its parameters.

Incheon. The coin-model tie probability 0.903% (which we re-verify) and the birthday-problem pair count are correct, but "suppose about 1% of pairs are similar" is an assumption. Measured (by inverting the same decomposition against our shell counts), the actual similar-pair rate is **0.012% — about 82× smaller**. The 158 early-voting units of Incheon contain just two near pairs at $d \leq 10$. The expected number of twins is therefore not the claimed 0.84 but 0.0137 (canonical) = 0.0146 (sideband), and $P(\geq 1) = 1.4\%$. By count magnitude (3,030/1,440), the Songdo twin is single-handedly the rarest twin in the country. In fairness, Songdo 1- and 2-dong are de facto parameter twins (4,546 vs. 4,539 ballots; homogeneous new-town districts), the one case where the demographic-similarity story is visibly plausible — favoring the original analysis on the *conditional* factor.

The follow-up addendum. The objection that assuming $p_1 = p_2$ inflates the tie probability is valid — it is the mirror image of our Proposition 1. However, the public code verification offered in response draws a single p **shared by both units** ($X_1, X_2 \sim \mathrm{Bin}(n, p)$ with one Beta draw), thereby re-assuming $p_1 = p_2$; its agreement with the original number (0.901% vs. 0.906%) is circular. Implemented as intended — each unit drawing p independently from its posterior — the variance triples and the tie probability becomes 0.524% (a $\sqrt{3}$ reduction). Either way the order of magnitude stands, so "the conclusion is roughly unchanged" holds in the narrow sense for the conditional factor. But the tie probability is extremely sensitive to the *true* difference in support: 0.54% at $\Delta\mathrm{p} = 1$ pp, 0.11% at 2 pp, 0.008% at 3 pp, ≈ 0 at 5 pp. The effective definition of "similar" is thus "true support within ±1.5 pp *and* nearly equal size" — and the measured fraction of such pairs is 0.012%. The thread polished the approximately-correct conditional factor while leaving the 80×-wrong unconditional factor untouched.

Gwangju–Jeonnam. The original analysis's qualitative directions — tie probability grows as n falls and $p \to 1$; pairs grow as k^2 — are all correct, and indeed explain why the Gwangju–Jeonnam expectation (1.47) is ~100× Incheon's (0.014). But no quantitative computation was performed before declaring the result "a naturally understandable coincidence." The measured significance is 1.0–2.8%: neither "entirely natural" (which would imply tens of percent) nor "an impossible coincidence" is supported.

7. Discussion

7.1 Interpretation

Three facts are established. (i) Twins are a base-rate phenomenon — the presidential election's seven twins match the model exactly. (ii) The nine twins of the 2026 local elections are a ~1% tail event relative to that base rate, a number stable across the entire methodological spectrum (conditional / unconditional / model-free). (iii) The excess is an atom at exact equality, localized four-fold.

What may be inferred? A frequentist 1% is not a posterior probability of fraud. The fraud hypothesis does not particularly predict this pattern — exact ties of tens of votes in districts decided by 60-point margins — so the likelihood ratio is near 1, and with a low prior and no independent evidence the posterior barely moves. Meanwhile the public scrutinizes dozens of statistics per election (selection effects [5]), so the effective surprise of a face-value 1% is smaller still. In the opposite direction, the rhetoric of "entirely natural" is, as a sampling claim, off by a factor of ~40 (§6). The sentence the data licenses is exactly this: **it is rare (~1%), and rarity is not evidence of manipulation.**

Two hypotheses remain for the residual atom: a post-hoc-selected coincidence, or a non-statistical cause confined to the 2026 local elections' early voting (including reporting-pipeline artifacts upstream of publication). The latter is discriminable by non-statistical means (§7.3) — a narrowing of the search space which is the product, not the failure, of the statistical analysis.

7.2 Limitations

(1) The fundamental limitation of post-hoc analysis of a single election: the hypothesis (the twin statistic itself) originated from the data. The presidential replication mitigates this by pre-specification, but a cause unique to the local elections survives it. (2) Each neighborhood is observed once, so within-neighborhood dispersion (φ) is unidentifiable. (3) The type localization (§4.3) is post hoc and its significance is reported at face value, before exploration correction. (4) The integrity verification (§2.4) excludes collection and transcription artifacts on our side, but errors upstream of official publication cannot be checked by re-reading the publication — that is the domain of physical recounts. (5) A replication on the previous local elections (2022) remains undone; it would discriminate "local elections in general" from "2026 only" and would sharpen, though not redirect, our conclusions.

7.3 Future work

(a) The 2022 local elections under the identical pipeline; (b) extension to National Assembly elections (constituency level); (c) generalization of the sideband estimator — arbitrary-dimension multivariate exact ties, shell weighting under heterogeneous noise scales.

8. Conclusion

The nine twin counting units of the 2026 Korean local elections constitute a tail event of roughly 1% under any smooth stochastic model (nationwide 0.64–0.94%; Gwangju–Jeonnam 1.0–2.8%). The figure is identical whether obtained from a generative model conditioned on observed parameters, an unconditional distributional model, or a sideband extrapolation using no generative assumptions at all; and the excess did not recur in the 2025 presidential election. Exact ties are a natural phenomenon present in every election; the 2026 excess lies within the range of rare but possible coincidence; and the final identity of its residue belongs to verification outside statistics — source audits and physical recounts. The error shared by both camps of the public controversy was the same: not measuring what is measurable.

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Appendix A. Reproducibility

All figures and numbers are reproduced by the public repository's scripts: `twin_model.py` (canonical model), `analyze_national.py` (local elections: national, shells, sideband), `analyze_president.py` (presidential), `bias_demo.py` (numerical verification of Proposition 1), `type_decompose.py` (type decomposition), `abinitio_sim.py` (unconditional variant), `verify_twins.py` (data integrity, §2.4), `paper/make_figs.py` (figures). Raw crawls and cleaned data are included.

Appendix B. Acknowledgements and AI disclosure

Anthropic's Claude (Fable 5) was used for analysis-design discussion, code authoring, and manuscript drafting. Responsibility for verification of all analyses and for the final conclusions rests with the author.